

Sample Slow-Problem Submission

Problem

A triangle is subdivided into smaller triangles. The three vertices of the large triangle are assigned three different colors, and the vertices along each side use only the colors of the two endpoints of that side.

Must there be a small triangle whose three vertices have three different colors?

Guidelines for Exploration

You are not necessarily expected to solve the problem immediately. As you work on the problem, consider the following questions and keep a brief record of your exploration.

1. What examples or special cases did you try?

Describe enough of them to show how you explored the problem.

2. What did you observe from these examples?

Look for patterns, exceptional cases, or features that seem important.

3. What pattern, rule, or general statement do your observations suggest?

If possible, formulate it as a conjecture, proposition, or possible theorem.

4. What approaches did you try?

Record both successful and unsuccessful approaches when they contributed to your understanding of the problem.

5. What mathematical progress did you make?

This may include a special case, partial proof, useful lemma, counterexample, invariant, counting argument, or an explanation of why an approach fails.

6. What would you try next?

Explain what questions or approaches seem most promising if you were to continue working on the problem.

7. Did you use any external sources or AI tools?

If so, briefly explain:

- what source or tool you used;

- when you used it;
- what you looked for or asked;
- what ideas, hints, examples, or arguments you obtained; and
- how you checked, modified, or developed them in your own work.

You do not need to reproduce an entire AI conversation, but any interaction that substantially influenced your mathematical work should be summarized.

Your submission should contain both an **exploration record** addressing these questions and a **mathematical write-up** presenting your current best understanding of the problem.

Part I. Exploration Record

1. Examples and Special Cases

I first considered the case in which the large triangle is not subdivided. In this case, the triangle itself has all three colors, so the statement is true.

I then drew several subdivisions with a small number of interior vertices and tried different colorings satisfying the boundary condition. In every example, I found at least one small triangle containing all three colors.

In the examples I tried, the number of such triangles was

$$1, \quad 3, \quad 1, \quad 5,$$

but never 0 or an even number.

2. Observations

I could not construct a valid coloring with no tricolored small triangle.

More interestingly, the number of tricolored triangles seemed always to be odd.

I also noticed that the boundary condition appears to be essential. If arbitrary colors are allowed on the boundary, the conclusion need not hold.

3. Conjecture / Possible Theorem

My first conjecture was:

Every valid coloring contains at least one small triangle whose vertices have three different colors.

After examining more examples, I arrived at the stronger conjecture:

Conjecture 1. *The number of tricolored small triangles is always odd.*

If this stronger conjecture is true, then the original statement follows immediately.

4. Approaches Tried

Attempt 1: Induction. I first tried induction on the number of small triangles. I hoped to remove one small triangle and apply the induction hypothesis, but I could not find a natural way to do this while preserving the boundary condition.

This made me think that ordinary induction on the number of small triangles might not be the most natural approach.

Attempt 2: Counting vertices. I also tried counting the number of vertices of each color, but these numbers did not seem to control the number of tricolored triangles.

Attempt 3: Looking at edges. I then looked at edges whose endpoints have colors 1 and 2. Along the side joining the vertices colored 1 and 2, the colors begin with 1 and end with 2. This suggested that the number of 1-2 transitions along that side should be odd.

I also noticed that every interior edge belongs to two small triangles, whereas every boundary edge belongs to only one. This suggested a parity-counting argument.

5. Mathematical Progress

I considered incidences

$$(T, e),$$

where T is a small triangle and e is a 1-2 edge of T .

A tricolored triangle contains exactly one 1-2 edge. A non-tricolored triangle contains either zero or two such edges. Therefore, modulo 2, only the tricolored triangles contribute to the incidence count.

On the other hand, each interior 1-2 edge is counted twice, while each boundary 1-2 edge is counted once. Thus the parity of the total incidence count is determined by the boundary.

Since the relevant boundary contains an odd number of 1-2 edges, the incidence count is odd. This appears to prove that the number of tricolored triangles is odd.

6. What I Would Try Next

I would write the parity argument more carefully, especially the claim that the number of 1-2 boundary edges is odd.

I would also like to investigate whether there are higher-dimensional versions of the statement and what happens if the boundary condition is changed.

7. Use of External Sources or AI Tools

After trying several examples and attempting induction, I asked an AI tool:

“What kinds of quantities might be useful to count in this problem? Please do not give a complete proof.”

The AI suggested considering edges whose endpoints have two specified colors.

I then examined 1-2 edges in my own examples and noticed the difference between interior and boundary edges. This led me to the incidence-counting argument above.

Later, I asked the AI whether my claim about the parity of the 1-2 boundary edges was correct. I used its response as a hint, checked the claim independently, and incorporated the argument into my own solution.

Part II. Mathematical Write-up

Theorem 1. *Under the given boundary-coloring condition, the number of small triangles whose three vertices have three different colors is odd. In particular, there is at least one such triangle.*

Proof Idea

Choose two colors, say 1 and 2, and count incidences between small triangles and edges whose endpoints have colors 1 and 2.

Interior edges are counted twice, whereas boundary edges are counted once. On the side of the large triangle joining colors 1 and 2, the number of 1-2 edges is odd.

On the other hand, a tricolored small triangle contributes exactly one 1-2 edge, while every other small triangle contributes an even number. Comparing the two counts modulo 2 gives the result.

Proof. Consider the side of the large triangle whose endpoints have colors 1 and 2. By the boundary condition, every vertex on this side has color 1 or 2.

Moving along the side from the endpoint colored 1 to the endpoint colored 2, the color must change an odd number of times. Hence there are an odd number of boundary edges whose endpoints have colors 1 and 2.

Now count incidences between small triangles and their 1-2 edges.

Every interior 1-2 edge belongs to exactly two small triangles and therefore contributes twice to the incidence count. Every boundary 1-2 edge belongs to exactly one small triangle. Consequently, the total number of incidences is odd.

For each small triangle, there are three possibilities relevant to this count:

- a tricolored triangle has exactly one 1-2 edge;
- a triangle using only colors 1 and 2 has either zero or two 1-2 edges;
- a triangle not containing both colors 1 and 2 has none.

Thus, modulo 2, the total number of incidences is equal to the number of tricolored small triangles.

Since the incidence count is odd, the number of tricolored small triangles is odd. Therefore at least one tricolored small triangle must exist. $\square$

Comments

1. The theorem obtained is stronger than the original existence question: it shows that the number of tricolored triangles is always odd.
2. The boundary condition is essential to the argument because it guarantees the required parity on the boundary.
3. The unsuccessful induction attempt was still useful because it suggested that a local reduction might not be the natural approach. The successful argument instead uses a global parity invariant.
4. The proof suggests possible generalizations in which an appropriate boundary condition forces a global combinatorial conclusion.